Citations

1. Number Theory: The Birch and Swinnerton-Dyer (BSD) Conjecture

This conjecture is the unproven foundation initially used as an axiom. The Clay Mathematics Institute (CMI) statement on the Millennium Prize Problem.

Concept to Cite	Source Description	Accredited Citation Source
Birch and Swinnerton-Dyer Conjecture (Formal Statement)	The official description of the Millennium Prize Problem, defining the link between the analytic behavior of the L-function at $s = 1$ and the algebraic rank of the Mordell-Weil group.	**Clay Mathematics Institute (CMI):** THE BIRCH AND SWINNERTON-DYER CONJECTURE (Source 1.4)
Rank Definition (Weak BSD)	Standard definition of rank, crucial for your **Weak BSD Equivalence** (File 2) and the predictive test (File 3).	**Wikipedia:** Birch and Swinnerton-Dyer conjecture (Source 1.3)

(This work augments this conjecture by providing empirical/predictive validation of the consequences of the conjecture in a physical domain (Files 2, 3, 10) and by demonstrating its structural rigidity (File 6).)

2. Foundational Geometry and Physics: Elliptic Curve Discriminant

The **Foundational Equivalence Hypothesis** (Files 13, 14) links the Discriminant (\Delta) to the Virial Energy Imbalance (|2T+U|).

Concept to Cite	Source Description	Accredited Citation Source
Elliptic Curve Discriminant (Δ)	Formal definition of the discriminant and its relationship to singularities (smoothness) and the Weierstrass equation. Essential for the Energy-Geometry Correspondence.	**Wolfram MathWorld:** Elliptic Discriminant (Source 2.3)
Conductor (N) / Discriminant Link	The relationship between \Delta and the Conductor N is critical for the **Modularity-Conductor-j-invariant (MCJ) Criterion** (Appendices).	**Scholar Commons:** The Relationship between Conductor and Discriminant of an Elliptic Curve Over Q (Source 2.2)

(This work augments these criteria by transforming the number-theoretic discriminant into the physical/thermodynamic metric, the Virial Discriminant Imbalance (VDI), used in File 17.)

3. Links to Fundamental Theory: The Langlands Program & Modular Forms

The **First-Principles Validation** (File 12) relies on linking cosmologically-derived curves to modular forms.

Concept to Cite	Purpose in This Work	Accredited Citation Source
Langlands Program Overview	Contextualizing the modularity link as a major set of conjectures relating number theory (Galois groups) to automorphic forms.	**Wikipedia:** Langlands program (Source 3.2)
Modularity Conjecture (Wiles' Work)	Citing the proven case (semistable curves) as the key connection between elliptic curves and modular forms (Source 3.4 is also relevant).	**DPMMS (Cambridge):** Elliptic Curves and Modularity (Source 3.1)

4. Methodology: Topological Data Analysis (TDA) in Cosmology

The **ACSC** (Arithmetic–Cosmic Structure Conjecture) and **ECC** (Entropy Cohomology Conjecture) are heavily dependent on TDA and Persistent Homology (Files 16, 17).

Concept to Cite	Purpose in This Work	Accredited Citation Source
Persistent Homology in the Cosmic Web	Citing the use of TDA and Betti numbers to analyze the topology of the large-scale structure (LSS), which is the target of Φ mapping.	**Monthly Notices of the Royal Astronomical Society (Oxford Academic):** Persistent homology of the cosmic web – I. Hierarchical topology in ΛCDM cosmologies (Source 4.5)
TDA for Parameter Inference	Citing the use of persistent homology outputs (persistence diagrams) with machine learning to constrain cosmological parameters, which is the exact predictive methodology of the ECC.	**arXiv:** Cosmology with persistent homology: parameter inference via machine learning (Source 4.4)

Concept to Cite	Purpose in This Work	Accredited Citation Source
Persistent Homology (Basic Theory)	Citing the foundational mathematical concepts of TDA, including the role of the Betti numbers (β_i) and filtration.	**Wikipedia:** Topological data analysis (Source 4.3)